Title: AI DRIVEN ESG ANALYSIS

Phase-2, Review-2

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ABSTRACT:

‘ Traditional ESG scoring methods are manual, subjective, inconsistent, and lack scalability. ’

Current ESG (Environmental, Social, and Governance) evaluation methods are predominantly manual, making them prone to subjectivity, inconsistency, and scalability challenges. This project aims to revolutionize ESG assessment and its application in investment strategies by introducing an AI-powered framework. Utilizing advanced technologies like Machine Learning (ML) and Natural Language Processing (NLP), the system automates the extraction of ESG data from both structured and unstructured sources.

This automation enables the generation of ESG scores that are more accurate, timely, and explainable. The framework is designed to enhance transparency and improve the predictive reliability of ESG assessments, ultimately delivering actionable insights to investors, regulatory bodies, and other relevant stakeholders. The overarching goal is to develop a scalable, efficient, and data-centric ESG evaluation model that promotes informed and sustainable investment practices.

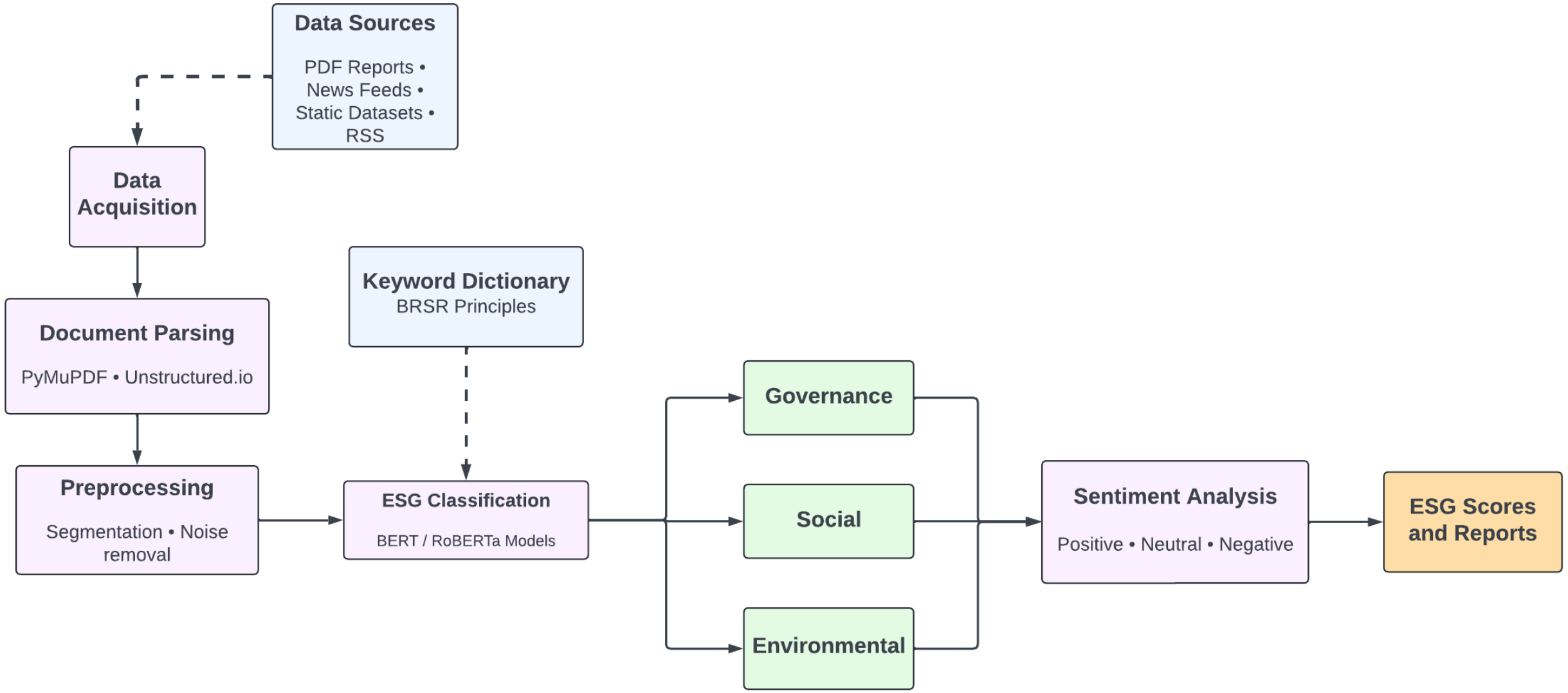
OBJECTIVES:

1. To develop a layout-aware ESG document parser that can extract text and tables from PDF-based sustainability reports.
2. To design and implement a multi-label classification model for categorizing extracted content into Environmental, Social, and Governance (ESG) segments.
3. To build and implement a sentiment analysis module that accurately identifies and categorizes the tone (positive, negative, neutral) of ESG-related content for each Environmental, Social, and Governance pillar.

EXTENDED LITERATURE SURVEY :

Paper / Source	Focus	Methodologies Highlighted	ESG Factors Considered	Unique Contributions
AI-driven ESG in Chinese SOEs	AI impact on ESG and sustainability	Regression models, text analysis of AI terms, composite ESG indices	Environmental, Social, Governance	Mediating role of ESG between AI and sustainability; detailed AI usage quantification
SAFE AI Metrics in Finance	AI models for ESG-based credit ratings	Machine learning models, SAFE AI metrics (accuracy, robustness, explainability, fairness)	ESG dimensions, regional and sectoral factors	Comprehensive sustainability evaluation metrics for AI models
S&P Global ESG Scores (2025)	Multi-level aggregation, question-level modeling, double materiality, industry-relative scoring	Environmental, Social, Governance with weighted criteria	Industry peer benchmarking, modeling for accuracy	Detailed hierarchical scoring system
LSEG ESG Scores (2025)	Percentile rank scoring, dynamic ESG magnitude matrix, industry and country benchmarking	10 ESG themes grouped into 3 pillars, material and comparable measures	Industry for E&S, country for G, dynamic weight adjustment	Dynamic weighting, large data set, outlier-resistant scoring

PROPOSED METHODOLOGY:



PROPOSED METHODOLOGY :

Normalization

For any field f:

$$\text{Normalized Value}(f) = \frac{\text{Value}(f) - \text{Min}(f)}{\text{Max}(f) - \text{Min}(f)}$$

(clamped between 0 and 1)

Subgroup Score

For subgroup g with n fields:

$$\text{Score}(g) = \frac{1}{n} \sum_{i=1}^{n} \text{Normalized Value}(f_i)$$

Pillar Score

For a pillar p:

$$\text{Score}(p) = \sum_{g=1}^{m} \text{Score}(g) \times \text{Weight}(g)$$

Final ESG Score

$$\text{ESG Total} = \sum_{p \in \{E,S,G\}} \text{Score}(p) \times \text{Weight}(p) \times 100$$

IMPLEMENTATION:

Normalization Metrics

```
{
  "Environment": {
    "renewable_energy_percent": { "min": 0, "max": 80 },
    "water_recycled_percent": { "min": 0, "max": 100 },
    "ghg_total_tco2e": { "min": 1000, "max": 500000 },
    "energy_consumption_total_mwh": { "min": 1000, "max": 100000 }
  },
  "Social": {
    "female_percent": { "min": 0, "max": 50 },
    "attrition_rate_percent": { "min": 0, "max": 30 },
    "training_hours_per_employee": { "min": 5, "max": 50 },
    "csr_spend_million_inr": { "min": 1, "max": 200 }
  },
  "Governance": {
    "independent_directors_percent": { "min": 0, "max": 100 },
    "board_size_count": { "min": 5, "max": 20 },
    "board_meetings_count": { "min": 2, "max": 12 },
    "whistleblower_complaints_resolved": { "min": 0, "max": 100 }
  }
}
```

Feature weights

```
{
  "Environment": {
    "scope1_tco2e": 0.10,
    "scope2_tco2e": 0.05,
    "scope3_tco2e": 0.05,
    "ghg_total_tco2e": 0.05,
    "ghg_intensity_tco2e_per_revenue": 0.05,
    "energy_consumption_total_mwh": 0.05,
    "energy_from_renewables_mwh": 0.05,
    "renewable_energy_percent": 0.15,
    "water_withdrawal_kl": 0.05,
    "water_consumption_kl": 0.05,
    "water_recycled_percent": 0.15,
    "waste_hazardous_generated_tonnes": 0.05,
    "waste_non_hazardous_generated_tonnes": 0.05
  },
  "Social": {
    "total_employees": 0.05,
    "female_percent": 0.15,
    "women_in_board_percent": 0.10,
    "women_in_senior_mgmt_percent": 0.10,
    "attrition_rate_percent": 0.10,
    "training_hours_per_employee": 0.15,
    "csr_spend_million_inr": 0.20,
    "csr_beneficiaries_count": 0.15
  },
  "Governance": {
    "board_size_count": 0.05,
    "independent_directors_percent": 0.20,
    "women_directors_percent": 0.10,
    "esg_board_committee_exists": 0.10,
    "audit_committee_exists": 0.10,
    "risk_committee_exists": 0.10,
    "board_meetings_count": 0.10,
    "whistleblower_complaints_resolved": 0.10,
    "regulatory_penalties_count": 0.05,
    "standards_and_certifications": 0.10
  }
}
```

IMPLEMENTATION:

Pillar Weights for Pharma

```
{
  "Pharma": {
    "Environment": 0.4,
    "Social": 0.3,
    "Governance": 0.3
  }
}
```

Scoring example of Sun Pharma


```
{
  "pillars": {
    "Environment": 65.92430555555555,
    "Social": 57.87333333333335,
    "Governance": 38.00000000000001
  },
  "total": 55.13172222222222
}
```

IMPLEMENTATION:

Groq API and model setup

```
# ------------------------------
# Setup
# ------------------------------
load_dotenv()
MODEL_NAME = "moonshotai/kimi-k2-instruct"   # change model if needed
client = Groq(api_key=os.environ.get("GROQ_API_KEY"))

# ------------------------------
# Global token counter
# ------------------------------
token_counter = {"prompt": 0, "completion": 0, "total": 0}
```

Schema Definition for strict JSON Extraction

```
# ------------------------------
# Schema (73 fields total)
# ------------------------------
SCHEMA = {
    "Environment": {
        "scope1_tco2e": None, "scope2_tco2e": None, "scope3_tco2e": None,
        "ghg_total_tco2e": None, "ghg_intensity_tco2e_per_revenue": None,
        "energy_consumption_total_mwh": None, "energy_from_renewables_mwh": None,
        "renewable_energy_percent": None, "water_withdrawal_kl": None,
        "water_consumption_kl": None, "water_discharge_kl": None,
        "water_recycled_percent": None, "water_intensity_kl_per_unit": None,
        "carbon_emissions": {"baseline_year": None,"target_year": None,
                             "target_reduction_percent": None,"achieved_reduction_percent": None},
        "water_consumption_reduction": {"baseline_year": None,"target_year": None,
                                        "target_reduction_percent": None,"achieved_reduction_percent": None},
        "waste_hazardous_generated_tonnes": None,"waste_hazardous_recycled_percent": None,
        "waste_hazardous_disposed_percent": None,"waste_non_hazardous_generated_tonnes": None,
        "waste_non_hazardous_recycled_percent": None,"waste_co_processed_percent": None,
        "nox_tonnes": None,"sox_tonnes": None,"pm_tonnes": None,
        "effluent_treated_percent": None,"sites_in_protected_areas_count": None,
    },
    "Social": {...},
    "Governance": {...},
}
```

IMPLEMENTATION:

Cleaning RAW txt file

```python
# ------------------------------
# Step 1: Process raw TXT file
# ------------------------------
def process_text(raw_path, cleaned_folder="cleaned_input_txt"):  1 usage
    """Clean raw text (remove headers/footers, normalize spaces)"""
    os.makedirs(cleaned_folder, exist_ok=True)

    with open(raw_path, "r", encoding="utf-8") as f:
        text = f.read()

    # Example cleaning (you can extend this as needed)
    text = re.sub( pattern: r"\s+",  repl: " ", text).strip()

    company = os.path.splitext(os.path.basename(raw_path))[0]
    cleaned_path = os.path.join(cleaned_folder, f"{company}.txt")

    with open(cleaned_path, "w", encoding="utf-8") as f:
        f.write(text)

    print(f"   ✨ Cleaned text saved at {cleaned_path}")
    return cleaned_path
```

Extracting BRSR Features

```python
# ------------------------------
# Step 2: Extract BRSR metrics
# ------------------------------
def extract_brsr_from_file(path):  1 usage
    with open(path, "r", encoding="utf-8") as f:
        text = f.read()

    company = os.path.splitext(os.path.basename(path))[0]
    print(f"\n➡️ Processing {company}")
    data = extract_brsr(text)   # this comes from your try_4.py

    os.makedirs( name: "extracted", exist_ok=True)
    out_path = os.path.join("extracted", f"{company}_metrics.json")
    with open(out_path, "w", encoding="utf-8") as f:
        json.dump(data, f, indent=2)

    print(f"   💾 Saved {out_path}")
    return out_path
```

IMPLEMENTATION:

Chunking and merging function

```python
# ------------------------------
# Utilities
# ------------------------------
def chunk_text(text, max_chars=4500):  1 usage
    return [text[i:i+max_chars] for i in range(0, len(text), max_chars)]

def deep_merge(base, update):  2 usages
    for k, v in update.items():
        if isinstance(v, dict) and isinstance(base.get(k), dict):
            base[k] = deep_merge(base[k], v)
        else:
            if v not in (None, "", []):
                base[k] = v
    return base
```

System Prompt

```python
# ------------------------------
# Extraction
# ------------------------------
SECTION_SYSTEM = (
    "You are an ESG/BRSR data extractor. "
    "Output ONLY strict JSON. "
    "All values must be numbers or null. "
    "Never output formulas or text. "
    "If unsure, output null."
)
```

IMPLEMENTATION:

Extracting features in JSON format

```python
def extract_section_json(section_name, section_text, sub_schema):  1 usage
    global token_counter

    schema_text = json.dumps(sub_schema, indent=2)

    prompt = f"Extract numeric ESG/BRSR values for {section_name}.\nSchema:\n{schema_text}\n\nText:\n{section_text}"

    resp = client.chat.completions.create(
        model=MODEL_NAME,
        response_format={"type": "json_object"},
        messages=[{"role":"system","content":SECTION_SYSTEM},
                  {"role":"user","content":prompt}],
        temperature=0,
        max_tokens=1500,
    )
```

```python
def extract_brsr(text):  1 usage
    global token_counter
    token_counter = {"prompt":0,"completion":0,"total":0}
    result = deepcopy(SCHEMA)

    for section in ["Environment","Social","Governance"]:
        print(f"    🔹 Extracting {section}")
        section_result = deepcopy(SCHEMA[section])
        chunks = chunk_text(text, max_chars=4500)
        for i, chunk in enumerate(chunks):
            print(f"      ➡️ {section} chunk {i+1}/{len(chunks)}")
            try:
                chunk_result = extract_section_json(section, chunk, SCHEMA[section])
                section_result = deep_merge(section_result, chunk_result)
            except Exception as e:
                print(f"⚠️ Error in {section} chunk {i+1}: {e}")
        result[section] = section_result

    result = postprocess_numbers(result)
    s = json.dumps(result)
```

IMPLEMENTATION:

```
# ------------------------------
# Step 3: Weighted Scoring
# ------------------------------
def compute_weighted_score(company_data, pillar_weights, field_weights, benchmarks):  1 usage
    results = {}
    total_score = 0.0

    # Go pillar by pillar
    for pillar, pillar_weight in pillar_weights["Pharma"].items():
        pillar_score = 0.0
        field_sum = 0.0
        weight_sum = 0.0

        # Loop through fields inside benchmarks for this pillar
        if pillar in field_weights:
            for field, w in field_weights[pillar].items():
                # Get value from extracted JSON
                value = company_data.get(pillar, {}).get(field, None)
                if value is None:
                    continue
```

```
                # Apply field weight
                field_sum += norm_value * w
                weight_sum += w

        # Average field scores (if no fields, score = 0)
        if weight_sum > 0:
            pillar_score = field_sum / weight_sum
        else:
            pillar_score = 0

        # Apply pillar weight
        results[pillar] = round(pillar_score * pillar_weight * 100, 2)
        total_score += pillar_score * pillar_weight

    results["Total"] = round(total_score * 100, 2)
    return results
```

Normalization and Scoring

```
# Normalize using benchmarks.json ranges if available
if pillar in benchmarks and field in benchmarks[pillar]:
    min_v = benchmarks[pillar][field]["min"]
    max_v = benchmarks[pillar][field]["max"]

    if min_v is not None and max_v is not None and max_v > min_v:
        norm_value = (value - min_v) / (max_v - min_v)
        norm_value = max(0, min(1, norm_value))  # clamp between 0-1
    else:
        norm_value = 0
else:
    norm_value = value  # fallback raw

# Apply field weight
field_sum += norm_value * w
weight_sum += w
```

OBSERVATION:

Company	Run 1 ESG Score	Run 2 ESG Score	Run 1 Risk (↓ better)	Run 2 Risk (↓ better)	Sustainalytics Risk	Risk Level
CIPLA	49.96	66.33	27.5	18.5	25.6	Medium
DivisLab	61.44	70.72	21.2	16.1	32.8	High
Zydus	47.51	47.51	28.9	28.9	29.2	Medium
Sun Pharma	57.64	53.37	23.3	25.6	24.0	Medium
LaurusLabs	32.13	32.13	37.3	37.3	33.7	High

Final score and Comparison against Sustainalytics Score

OBSERVATION:

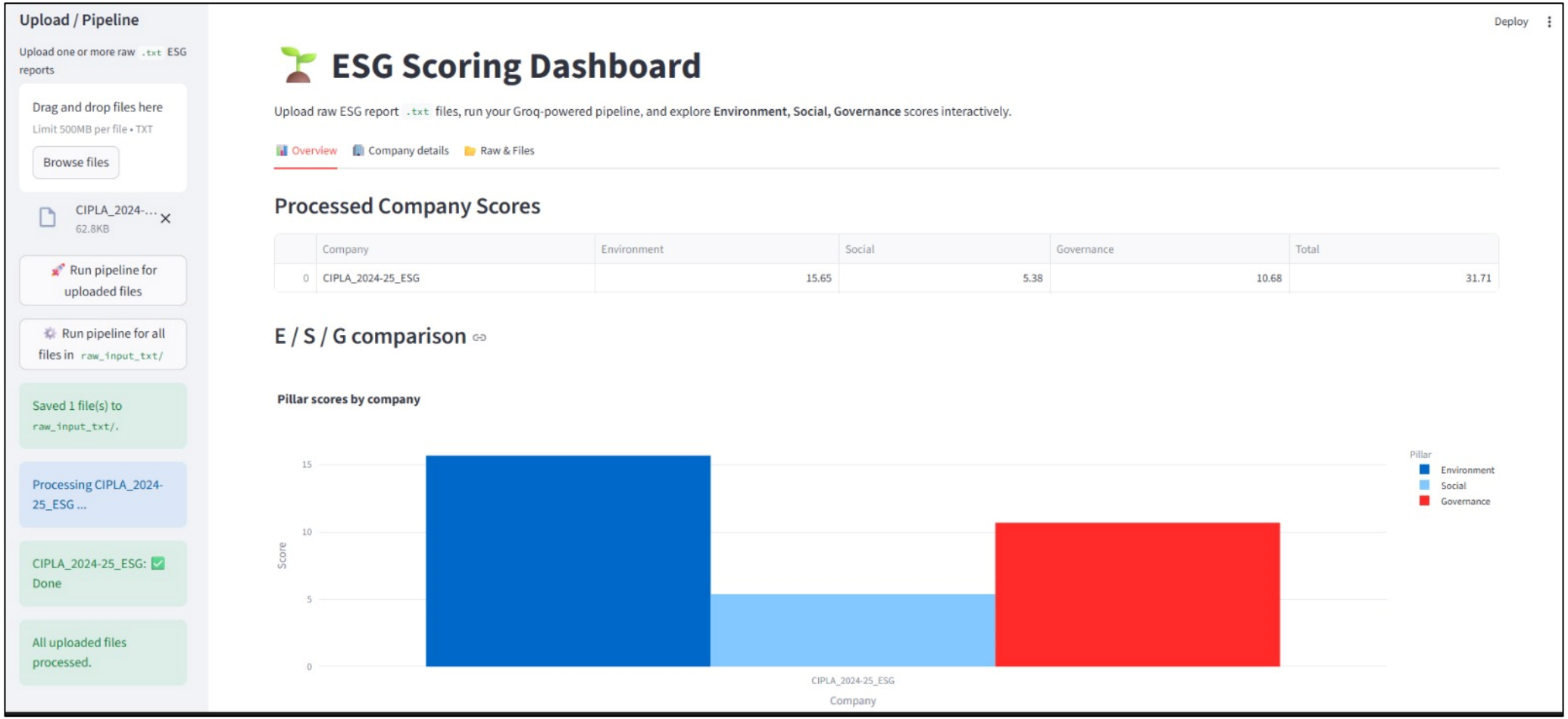
Runs performed using:

- **OSS-120B (Groq)**
- **DeepSeek-R1 (70B)**
- **LLaMA-3.3-70B Versatile**
- **Kimi-K2 Instruct-0905**

Sun Pharma Results by Model (Risk-Style)

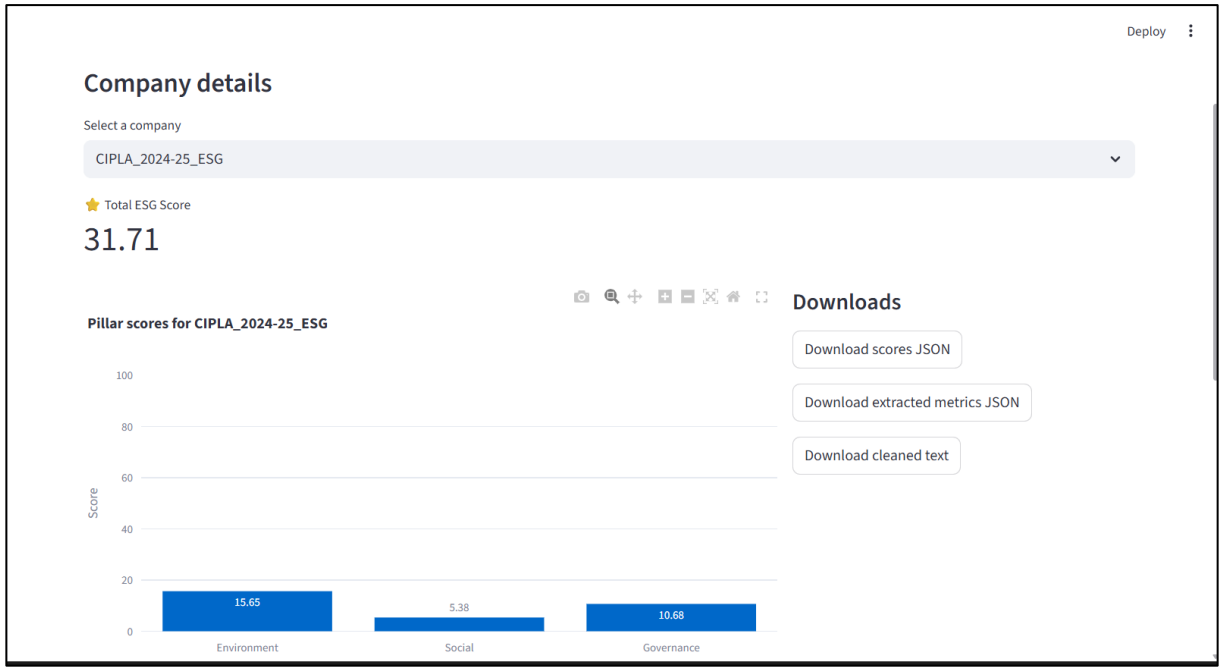
Model	ESG Score	Converted Risk Score	Pillar Weakness
OSS-120B	45.29	30.2	Governance partial coverage
DeepSeek-R1	45.48	30.1	Governance dropped → missing
LLaMA-3.3 Versatile	39.45	33.3	Social variability
Kimi-K2 Instruct-0905	55.13	24.7	More balanced, Governance extracted

SNAPSHOTS :

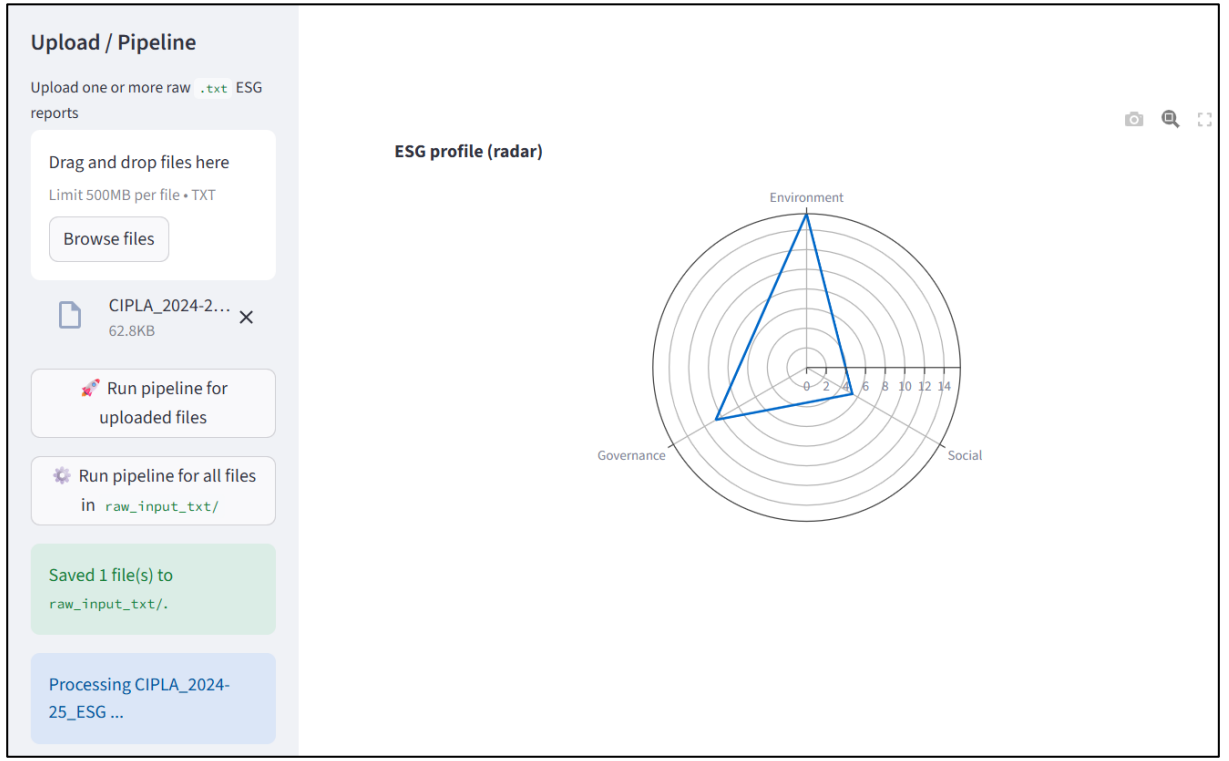


Dashboard Overview

SNAPSHOTS :



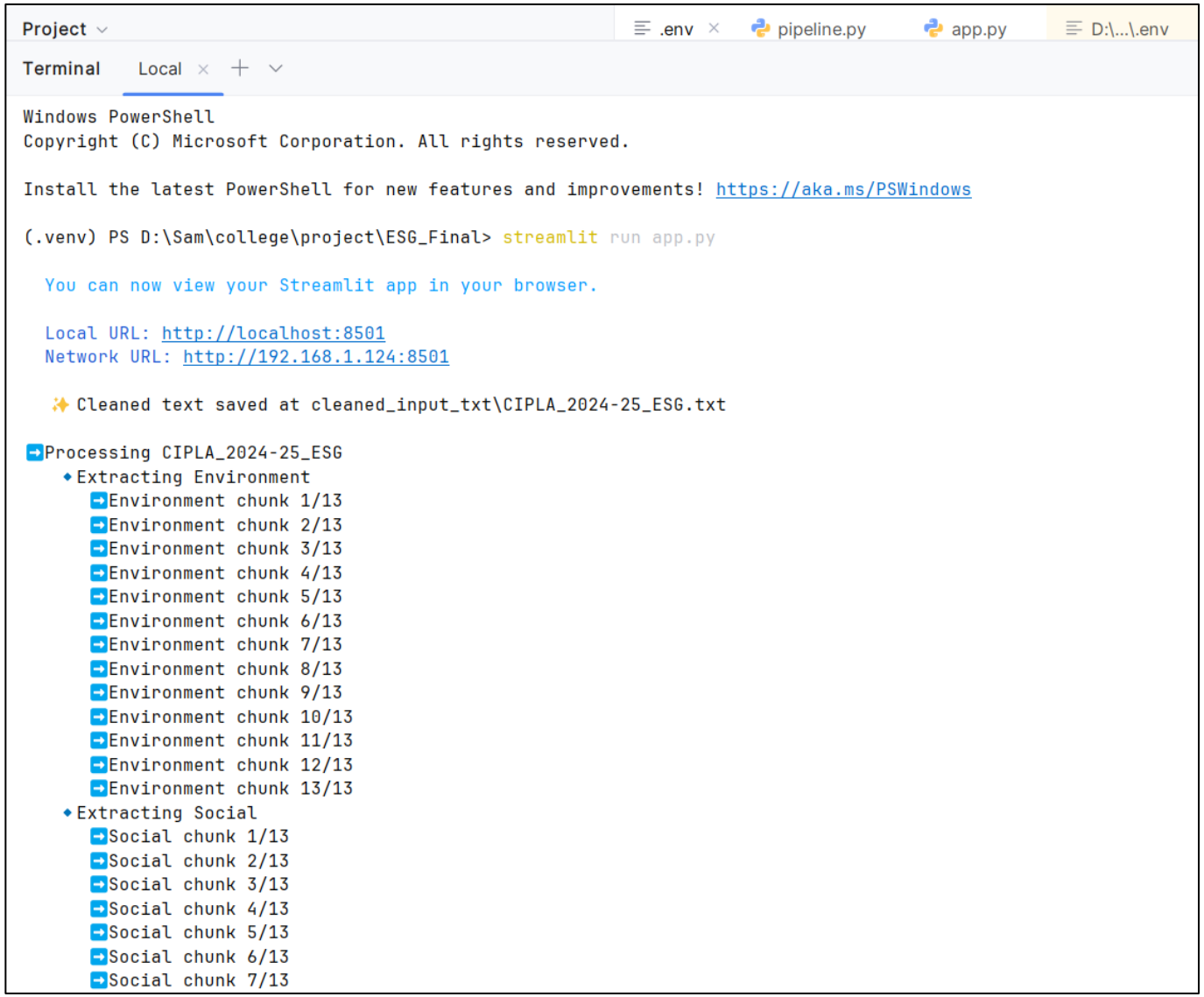
Company Details



ESG Profile

SNAPSHOTS :

Terminal Log



EXPECTED OUTCOMES:

1.Automated ESG Report Parsing :

Accurate extraction of text and tables from PDF-based ESG reports using layout-aware parsing tools.

2.ESG Category Classification :

Multi-label classification model to tag content under Environmental, Social, and Governance categories.

3.Sentiment Analysis Integration :

Polarity detection (positive/negative/neutral) for each ESG segment to assess disclosure tone.

4.Visual Dashboard :

A user-friendly interface for uploading reports and visualizing ESG classifications and sentiments.

5.Basic Model Explainability :

Use of tools like SHAP or LIME to provide transparency into model predictions.

REFERENCES :

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2. M. Singh, A. Kumar, and R. Gupta, "SAFE AI Metrics for ESG-Driven Credit Rating Models in Finance," *IEEE Transactions on Sustainable Computing*, vol. 10, no. 2, pp. 450–463, Apr. 2025.
3. S&P Global, "S&P Global ESG Scores Methodology," S&P Global, New York, NY, USA, Methodology Document, 2025.
4. London Stock Exchange Group, "LSEG ESG Scores Methodology," London, UK, Methodology Report, 2025.